

SWETANK KUMAR

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EDUCATION

SRM Institute of Science and Technology (SRMIST)

Expected 2029

B.Tech in Computer Science | Chennai, Tamil Nadu

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Web Development

EXPERIENCE

Open-Source Contributor – CERN HSF (GeneROOT / ramtools)

Jan 2026 – Present

Remote

- Implementing ramstats and ramsort genomic data tools using ROOT RNTuple for the GeneROOT infrastructure project under mentors from CERN and University of Plovdiv
- Developed Google Test suites for both tools; conducting compression analysis on 1000 Genomes Project SAM datasets to benchmark I/O performance of RNTuple vs TTree backends
- Opened GitHub issues documenting optimization proposals; collaborating asynchronously with international researchers in C++ and Python

Technical Developer – NVIDIA Student Development Program

Oct 2025 – Present

Remote

- Selected for NVIDIA's elite student program focused on GPU computing, CUDA programming, and AI acceleration on parallel hardware architectures
- Optimizing machine learning workloads for GPU execution and collaborating with student developers globally on high-performance computing projects

Software Developer – Team RUDRA (SRM Mars Rover)

Sep 2025 – Present

SRM Institute of Science and Technology | Chennai, Tamil Nadu

- Developing autonomous navigation and control systems for an international Mars rover competition; implementing FSM-based navigation using the YASMIN library
- Implementing computer vision algorithms for terrain mapping and obstacle detection using Python and OpenCV; contributing to embedded sensor integration and real-time data processing

PROJECTS & RESEARCH

MedBios – AI Clinical Report Intelligence Platform

2026 – Present

Python · OCR · Clinical NLP · Medical Knowledge Reasoning · LLM

- Building an AI-powered platform that ingests medical reports via OCR, applies clinical NLP to extract key lab values and abnormalities, and generates physician-ready structured summaries with evidence mapping
- Combines domain-specific medical knowledge reasoning with explainability features to ensure outputs are auditable and trustworthy in clinical workflows
- Designing the system with uncertainty-aware output layers to flag low-confidence extractions, supporting responsible deployment in diagnostic-adjacent tasks

Distributed ML Training Framework

2025 – 2026

PyTorch · Python · Distributed Systems · NCCL

- Built modular distributed ML framework implementing both Parameter Server and Ring All-Reduce synchronization backends with PyTorch autograd hook integration and pluggable communication layers
- Analyzed communication efficiency trade-offs between centralized and decentralized architectures, gradient staleness under asynchronous SGD, and scaling behaviour; validated with mock-based deterministic testing

CrowdAlpha – Multi-Agent Market Simulation Platform

Jan 2026 – Present

Python · Agent-Based Modeling · Limit Order Book · Monte Carlo

- Developed open simulation platform where market dynamics emerge from multi-agent limit order book interactions; demonstrated volatility clustering, spread dynamics, and tail behaviour from agent competition
- Engineered endogenous alpha decay model mapping strategies into a 6-dimensional factor space (turnover, holding period, directional beta, vol sensitivity, inventory skew, order aggressiveness) to quantify strategy crowding

Monte Carlo Pricing under Stochastic Volatility

2024 – 2025

Python · NumPy · Heston Model · Statistical Computing

- Implemented Heston stochastic volatility model for pricing European, Asian, and barrier options with control variates and antithetic sampling, reducing Monte Carlo variance by 60% while maintaining statistical rigor

Autonomous Pipe Exploration Robot

2024 – 2025

ROS · Arduino · Embedded C · LoRa · MQTT

- Engineered compact autonomous robot for confined pipeline navigation with LoRa-based telemetry and MQTT/Streamlit real-time data logging; achieved full mission completion in GPS-denied environments within 20-minute window

Neuromorphic Reservoir Computing with LIF Neurons

2024

Python · Nengo · Spiking Neural Networks

- Implemented Leaky Integrate-and-Fire neuron reservoir in Nengo for energy-efficient pattern recognition; published research findings on low-power edge computing applications

TECHNICAL SKILLS

Languages	Python, C++, C, JavaScript, SQL, Embedded C
ML / AI	PyTorch, Deep Learning, LLMs, Distributed Training, Neuromorphic Computing
Distributed Sys	Parameter Server, Ring All-Reduce, Async SGD, Gradient Synchronization, NCCL
Scientific	NumPy, SciPy, Pandas, Monte Carlo Methods, Stochastic Calculus, Agent-Based Modeling
Robotics	ROS, Arduino, LoRa, MQTT, OpenCV, Autonomous Navigation, YASMIN FSM
Web / Tools	MERN Stack, RESTful APIs, Git/GitHub, Linux, CMake, ROOT (RNTuple), LaTeX, CUDA

ACHIEVEMENTS & CERTIFICATIONS

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- International Rank 5 – CREST Olympiad (global problem-solving competition)
 - National Quiz Champion – Won 10+ national-level competitions including Inter APS Quiz (Runner-up) and USQ 2022 (Semi-Finalist)
 - Innov-athon 2025 – Certificate of Participation for innovative tech solutions
 - Certifications: Supervised Machine Learning (Coursera), C/C++ Programming (Coursera)